

CHAPTER 4

MASTERING CAR CONTROL

YOUR BIGGEST WORRY IN racing is losing control of the car. It comes slightly in front of "being bog slow."

And, being slow is bad enough, but being slow and out of control is worse.

To be competitive, you have to be willing, even eager, to flirt with the car's limits. The excitement begins when you turn the steering wheel, because you know that if the car isn't sliding you're probably not going fast enough. But once the slide begins, there is always a measure of doubt in your mind as to whether this slide is one that will go too far or is just another of the thousands of perfectly manageable ones that have gone before. With a veteran driver, one with good car control, the doubt is small, the confidence that every slide is controllable is high.

With a novice the doubt is greater, especially if you've spun in the recent past, and greater still if you don't know the reason for the previous spin. Without knowing why it happened and what you might do next time to keep the car under control, spins become random, unexplainable, events. It's like driving around with a bomb in the car that might go off at any moment—or might not.

GENUINE CONFIDENCE IS EARNED

Driving around the racetrack with this doubt hurts your performance in the car. You're going to be slower since you don't dare do anything that makes the car go faster, lest it fly off the road. Some would call this prudence and, while some people live quite happily with the motto "don't take a chance" they are not often successful racers.

This is not to say that cautiousness is all bad. A worse response to the anxiety is to pump up your confidence unrealistically—to use bravado to urge yourself to go faster. It doesn't work. If you don't have the skill required to control a sliding car, feigned bravery will only result in more spins, more loss of control.

Racers use steering, throttle, and braking to keep a racecar in control at its limit. Data collection can help pinpoint solutions to car control problems.

Confidence in your car control comes from having the experience of sliding the car and bringing it back from the edge of loss of control by

making the right moves. By practicing and getting good at it, even if you're inclined to be timid, you will be more willing to stretch the limits. If you're overly brave, you will do better by learning that it is skill that lowers lap time and not the willingness to sacrifice your car and yourself to the barriers.

"You've got to take small steps to find out where the limit is. You have to learn that fine line of where the car starts to break loose and learn to control it. But, to spin to find the limit? Not on a damned oval, you don't."

—Danny Sullivan

The "Necessity" of Spins

There is an attitude in the sport that spins—losses of control—are a necessary, even a beneficial, part of driving racecars. We strongly disagree. Any loss of control puts you in a position where what happens to you and your car is purely a matter of luck. With good luck the car spins down the middle of the road and you don't hit anything and nobody hits you and you drive away unscathed. With bad luck you hit something solid at high speed, and the results can be fatal.

Don't kid yourself. Although motor racing has gotten much safer in the last decade, you're driving lightweight vehicles at high speed and bad things do happen. Bad things happen less often to drivers who maintain control of their cars.

When it comes to driving faster than the rest, under complete control, it would be good to use Rick Mears as a role model. When Rick retired in 1993 he had run 15 years at Indianapolis, qualified on the pole (faster than 32 to 40 of the country's fastest drivers) six times, was on the front row eleven times, won the race four times, and, in all that time, spun and hit the wall

twice—both times because of a mechanical failure. Spinning is not necessary for success—just ask Rick.

Spinning is evidence of failure to accomplish the most basic role of a race driver: keeping the car on the road.

"I just don't think spinning the car trying to find the limit works. How do you measure the limit? If you're going backwards into the wall you've exceeded the limit, okay, but what do you learn from that?"

"I understand how it feels in the car when you've overstepped the amount of traction or grip you've got. That's how I know you've passed the limit. I don't get what revolving gets for you. I don't get it. Some guy goes flying off the road, then says 'Well, that's too fast.' What's he do, go back and look at where the speedometer needle stuck on impact?"

—Skip Barber

Finding the Limit

Some would say, "You can't find the limit if you don't go over it from time to time and spin." Nonsense.

You find the limit by taking small bites, experimenting with a little higher entry speed, a little earlier throttle application. Carrying more speed on the same arc will create a greater angle of slide, which might be faster, or might scrub off speed, making you slower. You don't need to spin to find this out.

Perhaps more entry speed makes the car wander wide of the best line and you need to give up some throttle on the exit of the corner to keep the car on the course. Again, you found the limit without losing control. When your car control is good you don't worry so much about spinning as you do about doing something in the corner that will cost you speed. Control is never in doubt.

"The mistake, going over the limit, makes the car go too far sideways and you correct it, and the car comes out of the corner slower. You blew the corner, but not the car. To lose it you're taking too big a bite, or you don't have the skill to recognize a mistake and fix it. You've got to know you have a problem then know how to solve it. If you have to be going backwards to know you made a mistake, you're not too smart."

—Skip Barber

REDEFINING OVERSTEER, UNDERSTEER, AND NEUTRAL

In our chapter on the three basics we introduced the fact that cars cornering at the limit of adhesion can do so with three different attitudes: understeer, oversteer, and neutral. In understeer, the front tires reach their cornering limit before the rears; in oversteer, the rear tires reach the limit first. In cornering a neutral car at the limit, both front and rear tires operate at their cornering limits and the car as a whole, since all four tires are contributing their fair share of the cornering load, generates its best cornering force.

Each of our illustrations of understeer, oversteer and neutral steer had one thing in common: the nose of the car was pointed in a direction *inside* the line defining the arc the car was traveling on.

Yaw Angle

When the car is pointed in a different direction than its direction of travel it is developing "yaw." Yaw is a phenomenon we see whenever cars change direction, whether it's at the cornering limit or in everyday street driving. Sometimes this yaw angle is relatively small, especially when dealing with low cornering forces, but sometimes, as with sprint cars on a dirt track, the car is pointing in an entirely different direction than the direction it is traveling. This can be entertaining to watch, and either exciting or terrifying when you're behind the wheel.

In Fig. 4-1, the car is turning right on an arc with a 100-foot radius. At any given instant its direction of travel is represented by a line tangent to the arc at the car's center. This line is called the car's "instantaneous direction of travel." The "yaw angle" of the car in this diagram, 10 degrees, is simply the angle between the direction of travel and the centerline of the racecar.

While it's impressive to watch cars being driven at large yaw angles, racers need to use just enough yaw angle to help the car go faster; spectacular slides, although they look so, aren't necessarily fast.

Slip Angle

"Slip angle" is the direction the wheel rim is pointing versus the direction the tire is traveling. The simple distinction is that slip angles refer to tires while yaw angle refers to the whole car.

Take a look at Fig. 4-2. This race car is cornering at its limit on a 100-foot radius arc. At the particular instant that this illustration captures, the driver has the steering wheel straight ahead. It might have been turned more to the right an instant before, or it might have been turned more left—not important. Now it's dead straight.

Let's draw a line in the center of each tire representing the direction each wheel rim is pointing. Here we'll say that each wheel is aligned perfectly with the

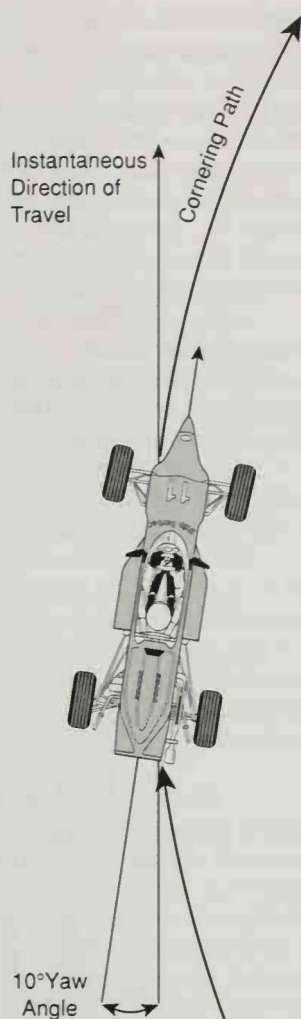


Fig. 4-1. A vehicle's yaw angle is the angle between the direction it is pointing versus the direction it is traveling.

centerline of the car, although with real cars this is often not the case due to toe-in or toe-out of the wheels.

To measure the slip angle of the tires, compare the direction the wheel is pointing to the direction the tire is traveling. The direction of travel of the tire is a line tangent to the 100-foot arc of the corner *at the tire*.

When you measure the difference between the tangent (the direction it is traveling) and the centerline of the wheel rim (the direction it is pointing), you find that the slip angle of the rear tires is 12 degrees.

At the front, the slip angle turns out to be eight degrees. It is slightly different from the rear because the front tires are eight feet further around the corner, and so the tangent lines differ by four degrees.

You can also see from this illustration that if you really wanted to be accurate about the slip angle of

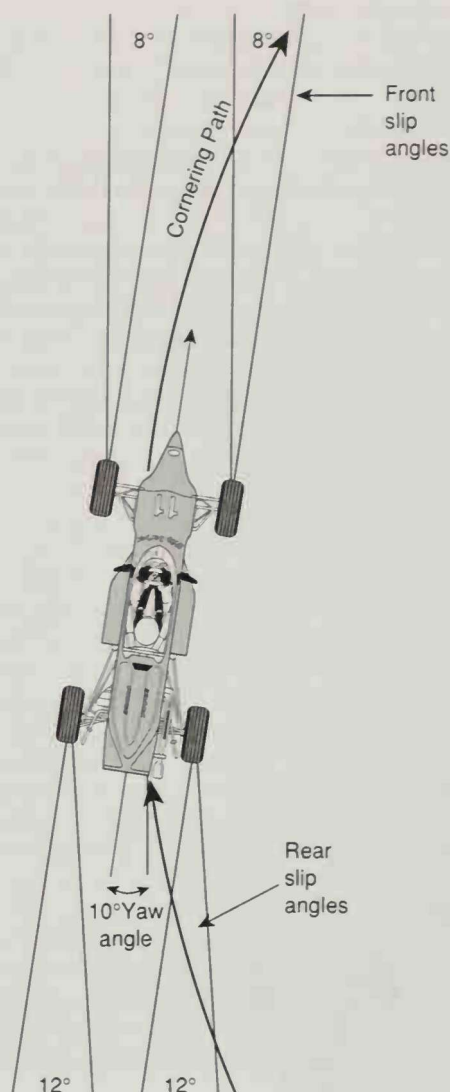


Fig. 4-2. "Slip angle" refers to the direction the wheel rim is pointing versus the wheel's path over the surface of the road.

each tire, you would have to draw radius lines to each tire's contact patch and take any toe changes into account in your measurements.

For our purposes, it is enough to know that slip and yaw are different and that, for a given amount of yaw, the front tires need to add angle to the inside of the turn to match the yaw angle, and that the slip angle of the rear tires is slightly larger than the yaw angle.

Slip angle also varies with corner radius. The deflection of the fronts inward needs to increase as the radius gets smaller and decrease as the radius gets larger. The slip angle of the rear tires will be proportionately larger than the yaw angle in tight radius turns than it will be in large radius turns.

Redefining Attitude

Up to now we defined understeer and oversteer in terms of which end of the car “slides” first. While this is an okay conceptual way of describing the sensation, it’s not really true. A car doesn’t need to slide to exhibit understeer or oversteer. A tire doesn’t necessarily have to be at its cornering limit to encounter slip angles.

Even at low speeds and cornering loads, cars develop slip angles at the front and rear tires. Consequently the car as a whole develops a yaw angle. At low loads the slip and yaw angles are small, but they’re there. A more accurate way of defining a car’s cornering attitude at both low speeds and high is to compare the slip angles of the front and rear tires.

New Definition of Understeer

If the slip angles of the front tires are greater than the slip angles of the rears, the car is understeering. The sensation you’ll get is that, relative to previous experience, the front wheels are pointing further into the corner than they should be considering the direction the car is pointed. In Fig. 4-3, the first car is in a classic state of understeer. The slip angle of the front tires is at 14 degrees, and the slip angle of the rear tires is at 6 degrees. The overall yaw angle is a modest 4 degrees.

New Definition of Oversteer

In the middle of Fig. 4-3, the same car is oversteering. The slip angles of the rear tires is at 16 degrees, the yaw angle of the car is at 14 degrees and, typical of this

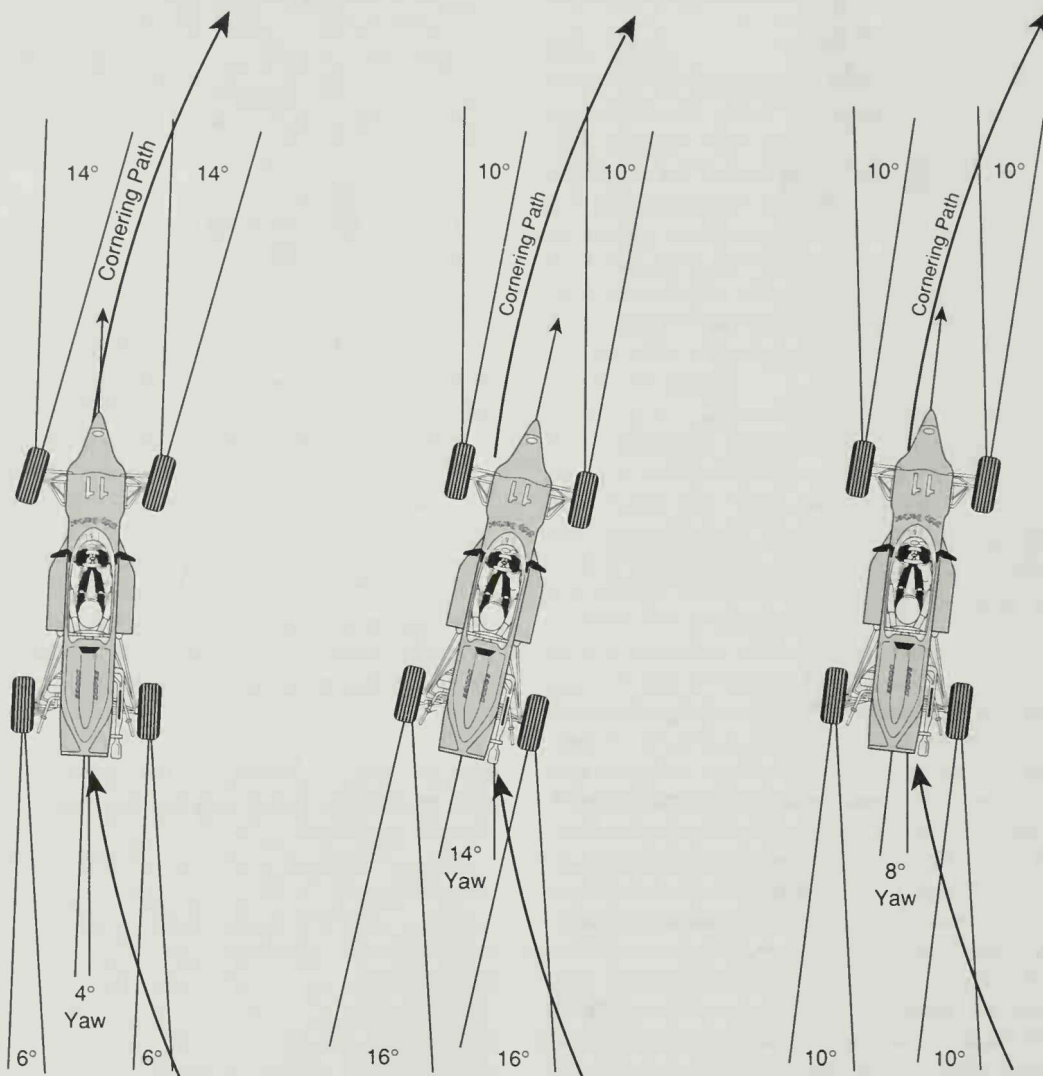


Fig. 4-3. An understeering car will have larger slip angles at the front tires than at the rear. An oversteering car will have larger slip angles at the rear tires than at the front. A neutral handling race car will corner with all four tires operating at the same slip angle. For maximum cornering force this slip angle will be in the range where the tires create maximum grip.

oversteer attitude, the front wheels are turned slightly left with a slip angle of 10 degrees.

Neutral Yields the Best Grip

Of the two cornering attitudes, the oversteering car certainly *looks* faster than the understeering car. The best cornering speed, however, doesn't come from either of these setups.

In the chapter on tires, you'll find that the ultimate cornering grip available from a tire comes in a narrow range of slip angle. The best slip angle will vary from tire to tire and car to car but, for now, let's say that you get the best traction when the tires operate at a 10 degree slip angle. In Fig. 4-3, the car with the most total grip, and consequently the highest potential cornering speed, is the car on the far right. The front and rear tires are operating at 10 degrees of slip, at their peak cornering traction, and since the slip angles are the same, the car is said to be "neutral."

Remember, though, that the car is neutral but it still has some yaw. Many drivers mistakenly think that a neutral car simply tracks around the corner with zero yaw. It doesn't. Neutral simply means that the front and rear slip angles are matched, hopefully in the range where the tires deliver the most traction.

Cornering Balance Is Fluid

A frequent mistake is to use one of these terms to sum up a car's cornering behavior. Pronouncements like "Porsches oversteer," or "front wheel drive cars understeer," or "mid-engine cars are neutral" are too generalized. The same car can exhibit understeer, oversteer, and neutrality under different conditions, often in the time it takes to drive through one corner. The process is a fluid one where you get a feeling for the right vehicle attitude and manipulate the controls of the car to keep cornering with just the right amount of yaw and slip.

USING THE CONTROLS TO ALTER HANDLING BALANCE

The key is that, although cars have designed-in tendencies toward oversteer, understeer or neutrality, you can change the car's characteristics by using the controls in different ways.

It's always dangerous to generalize, but the following guidelines show the handling effect of different control actions. These rules are for the most common racecar arrangement, rear wheel drive. Front wheel drive cars have some special handling considerations and we'll take a close look at this type of racecar later.

Braking Control

First, let's look at the corner entry, the portion of the turn between the turn-in point and the throttle application point.

"I think that the level of slide you use, the amount of yaw that works best, is specific to the car. For example, on a Formula Ford with street radial tires you can get away with a lot of slip angle in the tire without scrubbing much speed, but if you were to put slicks on the car you would have to use a lot less slip angle, or yaw, through the corner."

"I think that the tire and the amount of grip determine that. But I haven't changed what I do much. I never used big angles on the car—I just don't throw the car around a lot. I've stuck with that approach and I think it's been more of an advantage as I've moved up."

"There are examples of guys who have driven cars crossed up all the way through their careers—Gilles Villeneuve, for example. It's exciting to watch and sometimes I wish I drove more that way because it makes you look spectacular, but I don't think a lot of good things come out of that. You're using your tires more, you're heating your tires more and you're putting the car in an unnatural position. Especially with a ground effects car, when you start changing the angle of attack of the car to the airstream, you start losing downforce. Once you start to slide and you're losing downforce too, you're just compounding your own problem."

"In an Indy Car, unless it's a very slow corner and the driver's got wheelspin at the exit, you very rarely see big angles on the car. It doesn't help the car go fast."

—Bryan Herta

A properly biased braking system should have a neutralizing effect on the balance of the car past the turn-in point, reducing the cornering traction of both front and rear tires proportionately. Experience has shown, however, that this general rule can vary widely. Cars with slightly too much front brake bias will tend to understeer at corner entry while cars with slightly too much rear bias will tend to oversteer.

In either case, if the relaxation of brake pedal pressure is gradual; the balance you had when you first turn in will continue to the throttle application point. A snap off the brake pedal will make the car's balance go toward oversteer. Any pause between the brake release and the throttle application will also move the balance toward oversteer since the pause essentially causes trailing throttle oversteer.

Throttle Control

At the corner exit, where you're trying hardest to accelerate the car away from the corner and onto the

following straight, gradual applications of throttle tend to create understeer. Abrupt throttle applications, especially with powerful cars and/or limited grip, tend to create oversteer. Lifts off the throttle change the handling in the direction of oversteer. The more abrupt the lift, the more dramatic the effect.

"The tires give you the G and it's the driver's job to find how much pedal—either one—he can put down at the time for what he's asking the car to do. It's like asking from second to second, 'Can you take this?'"

"In real life we have to make up for our errors by adjusting the controls to keep the car on the racing line at close to the best angle."

—Terry Earwood

Steering Control

Done perfectly, your initial turn of the steering wheel at the turn-in point should initiate a rotation of the car around its center to the point where the vehicle's overall yaw angle is in the range of the tires' optimum slip angle. Ideally, once the car is at this angle, you barely move the steering wheel again as you maintain this perfect angle using the braking and turning ability of the car on the way into the corner, and the accelerating and turning ability on the way out. It can actually happen this way. Honest! Every now and then you do it just right, and it's the stuff grins are made of.

Often you use both the throttle *and* the steering wheel to fix a car control problem. Often you use the brake *and* steering wheel together as well.

ROTATION

As a car enters a corner, it is turning, that is, changing the direction it is traveling, but it is also rotating: as the car goes from zero yaw to whatever yaw angle it's going to operate at in the corner, there has to be some rotation around its center.

Let's use a model to make this phenomenon more clear. First, think of a vehicle that would corner with zero yaw. The best example is a railroad train negotiating a turn. When it is doing so, the centerline of the train and each of its cars would be perfectly aligned with the direction it's going at any instant. As the train enters a corner, it changes direction, but it doesn't develop any yaw (see Fig. 4-4).

Cars are different. To picture how a car both changes direction and rotates into yaw, let's mount a lazy susan on one of the train flat cars and put a race-car in the center of the lazy susan.

If the car were taking the corner without any yaw it would stay perfectly aligned with the train's centerline. Since all cars develop some yaw when cornering,

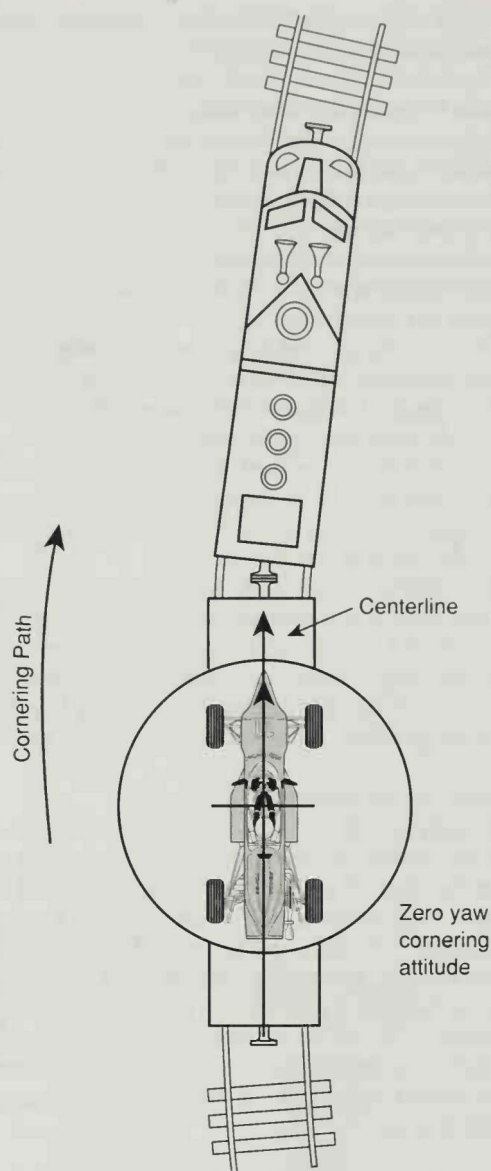


Fig. 4-4. Cornering with zero yaw is akin to taking a corner on rails.

however, the car on the lazy susan will rotate clockwise some time after the turn has begun.

In our race car example we want it to rotate 8 degrees, up to the range where the tire slip angles will be in the area where they make the most traction. At the exit of the corner, the rotation back to zero yaw will be back 8 degrees counterclockwise, (see Fig. 4-5).

This rotation has magnitude (the size of the angle we rotate to) and velocity (how fast the car rotates).

Your aim is to rotate the car to the best yaw angle at a rotation speed that makes it possible to tightly control the yaw. Cars that rotate fast are typically harder to control than cars with a slower rotational speed.

The struggle is to not over-rotate or under-rotate. Rotation, done right, is a thing of beauty. As the steer-

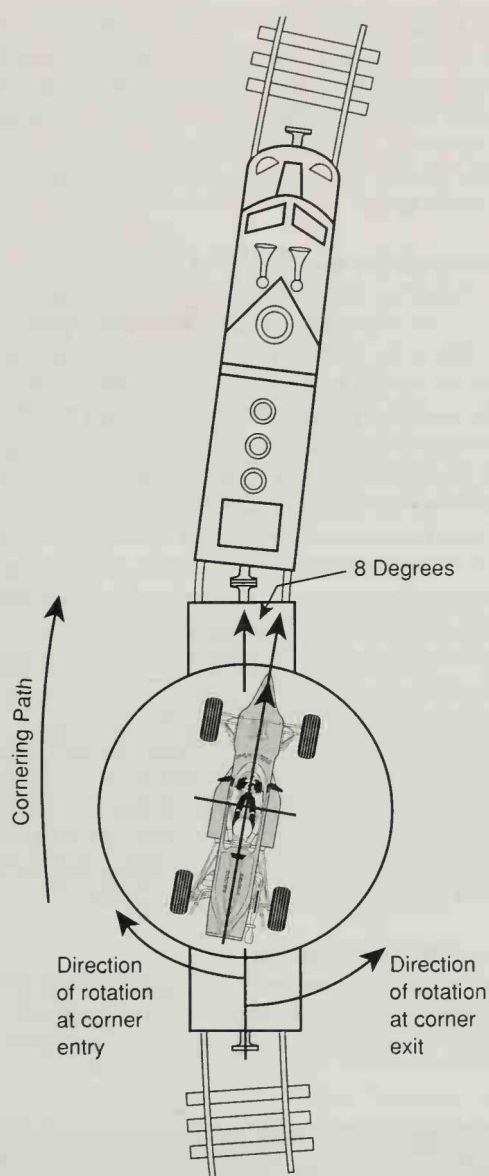


Fig. 4-5. At corner entry the car not only changes direction, it rotates around its center, creating a yaw angle. At corner exit, the rotation is in the opposite direction as the car goes back to zero yaw entering the following straight.

ing wheel is turned, the car not only changes direction, it rotates up to its best yaw angle in a gradual, manageable fashion. The initial steering input frequently has to be adjusted to maintain the proper angle as does the pressure on the brake pedal or throttle.

The control you adjust and the amount of the change depends on whether the car has over-rotated or under-rotated and where in the corner the under-steer or oversteer occurs.

OVER-ROTATION

Your aim is to allow the car to slide up to some optimum yaw angle, then hold it there for the duration of the corner if you can. By doing so, you'll be operating the car in a range where the tires deliver their maximum grip; more grip allows you to raise your cornering speed, the ultimate aim of car control. Let's go a step beyond theory and look at some real-life car control problems you'll face when you get going quickly on the racetrack.

Over-Rotating the Corner Entry

On the entry to Turn 7 at Sebring, you turn in a tad too late and pop off the brake pedal just past the turn-in point. The net effect is that the car not only rotates around its center the 7 to 10 degrees it should for optimum grip, it continues to rotate 10 degrees more into severe oversteer (see Fig. 4-6). The nose of the car is pointed inside the apex and if you don't do something fast, the car is going to continue rotating until it arrives at the apex backward.



Fig. 4-6. By suddenly releasing the brakes at the turn-in point you create a rotation bigger than needed to maximize the tires' grip.

Sensing the Oversteer

The good driver is already looking at the apex when the steering wheel is first turned, and the first indication of oversteer is the recognition that the nose of the car is pointed too far in for this stage of the corner. The other indicator is the seat of the pants feel, the kinesthetic sense that the motion of this car around its center is too rapid.

Correction

The first reaction is with the hands—dial in opposite lock, steering in the direction that the rear end of the car is trying to go. This is the "correction" phase of skid recovery.



Fig. 4-7. After correcting with opposite lock steering and settling the rear of the car with a subtle application of power, the “pause” at the peak of the slide is the trigger to recover by getting the steering wheel back toward straight.

Settle the Rear

The second, almost simultaneous action should be to settle the rear of the car by transferring weight onto the rear tires. You do this with throttle application—a *little* throttle application. A common mistake here would have been to instinctively jump on the power to save the slide. If you did that, you would have *lost* cornering traction at the rear by asking for too much acceleration traction.

By squeezing on the power, applying 30% throttle or so, you move load to the rear and cancel the engine braking effect of the abrupt snap off the brakes.

The Rotation Stops

The car begins to recover. The rotation around the center slows, then stops from the steering wheel correction and the throttle application. You’re in the eye of the storm. The rotation has paused at its peak. This is the “pause” phase, your cue to start taking out the opposite lock.

The Need for Recovery.

Now the rotation starts back the other way, counterclockwise, reducing the car’s overall yaw angle. If you don’t take the left steering out fast, you’ll get caught with the car back at zero rotation, a lot of load on the right tires and the steering wheel turned left. You’ll have saved the clockwise slide but gotten caught by the reaction of the rear rotating counterclockwise—you’ll end up spinning in the opposite direction.

This time you do it right and quickly get the wheel back to center following the pause: you successfully dissipate the rear end’s momentum back toward the right. The car settles down with an overall yaw angle less than you would have liked, but you didn’t spin.

Back to Business

As it settles you take the throttle on toward the floor, feeling just how much it will take without getting the rear out again. Total time of the event: 2.1 seconds from turn-in to recovery. Total time lost relative to your best job through Turn 7 is .15 seconds slower, with a corner exit speed 1 m.p.h. lower.

Saved But Not Forgotten

You saved the situation on this pass by using good car control skills, but you don’t want to do this on every corner. First, it’s slower, and second, you don’t want



Fig. 4-8. The car is back to a sensible yaw angle at the apex and what might have been a spin is merely tenths of seconds lost on this particular lap.

to go to the well too often. The bucket only has to come up empty once.

Driving away from Turn 7, a good driver would be doing a quick analysis of what went wrong so that something different can be done on the next lap to avoid the mistake: in this case, turn earlier and relax the braking pressure more slowly so that the car doesn't over-rotate.

There's pressure to squeeze in this self-analysis in a short period of time. You don't want to be preoccupied with the Turn 7 error as you enter Turn 8. We see it all the time. A driver has a "moment" in a corner, gets rattled or preoccupied with the mistake, and goes off in the next turn. You have to put the mistake behind you. Try to figure out what went wrong, when time allows, but crank up the concentration for each corner. Of course, the pressure and the potential for choking vary widely.

"It's always the case that when you make a mistake, it's over, it's done with, you have to put it behind you. It's hard to do. I actually find it easier to do if I'm racing with someone rather than alone. Little mistakes are like warning shots for me: 'That was a close call—you've got to pay attention—be alert.'"

—Danny Sullivan

Choke Etiquette

If this were a practice session, the amount of time you lost would be no big deal. But on the last lap of a race with a competitor on your gearbox, it's a whole different story. The frustration level can go through the ceiling when, through a simple mistake, you've given away a race.

This situation calls for emotional self-discipline. You've got to concentrate on driving and avoid the temptation to punch the steering wheel or hit yourself in the helmet. There is always time in the race to do something even dumber than you just did, and you don't want your agitation to get you into another mistake. You have to remind yourself that it could have been worse. You could have spun or crashed. You could have been running 14th. Try to be mature, at least until you park the car in the pits. Then you can throw your gloves at the dashboard and kick rocks around the paddock.

A Closer Look at Over-Rotation

In this example the causes of over-rotation were a late turn-in coupled with a violent snap off the brake pedal. Either could have caused the oversteer, but the pop off the brake pedal guaranteed that oversteer would be the end result of the late turn-in.

"If someone hits you and puts you off, you get really pissed. If you spin by yourself and miss your opportunity, then you're pissed at yourself. One is more anger, the other is more frustration. But either way, you have to take the energy you get from that and focus it on driving the car."

—Jeremy Dale

In a late turn-in, you sense that it's going to be tough getting to the apex, so you instinctively turn the steering wheel more than normal—and more abruptly than normal. What makes the situation worse is that the late turn-in requires an arc of a smaller radius than the proper turn-in; a tighter arc requires lower speed.

If you were ever going to be on or over the limit, this is it. The pop off the brake pedal suddenly relieves the front tires of any braking loads, and since the front tires are still heavily loaded from the weight transfer, they have gobs of cornering power. The snap off the brake also leaves the rear lightly loaded and high in the air. To top it off, unless you go immediately to some power, the engine compression acts as a rear-end brake, which, coupled with the rest, guarantees a sudden rotation toward oversteer.

This certainly is not the only way to get oversteer on the corner entry. Trailing throttle will tend to create many of the same problems: there's load up front to help the front bite, an unloaded rear, and drag on the rear tires from engine compression.

A sudden turn-in can also provoke over-rotation. The load transfer off the inside tires to the outside has an element of impact rather than a smooth transfer. This speeds up the rotational velocity, giving you less time to catch the rotation at a reasonable yaw angle.

A mismatched downshift will really get the rear out, especially at or after the turn-in point. If you chirp the rear tires just when rear end traction is needed for the correct handling balance, the tail may come out too fast for you to catch.

There are also causes other than improper technique. The front-to-rear handling balance of the car, whether it be chassis related or aerodynamic, might be too much in the direction of oversteer. Or perhaps the problem involves too much rear brake bias reducing the cornering grip of the rear tires.

Whatever the reason for the over-rotation, correction, pause, and recovery define the steering solution and transferring load with light throttle handles the footwork.

Over-Rotation by Power Oversteer

Sometimes the entry is just fine but something happens on the way out that gets the car sliding more than you want it to. For example, in your eagerness to be-

gin your run onto the straight as early and as forcefully as possible, you might open the throttle too soon or too abruptly. Let's take a look at Turn 9 at Sebring, the carousel leading onto the longest straight.

On this lap your approach to the corner is routine, decelerating down to about 70 at the turn-in. You continue to decelerate using braking and turning down to 63 m.p.h. at the throttle application point, then you're on it hard, going from zero throttle to full throttle in 30 feet. There was adequate rotation on the

way into the corner but the sudden application of throttle kicks the rear end out to the right, threatening to spin the car.

Correction

Your first reaction is to turn the steering wheel right to try and get the rear back in line. It works at first and you don't lift the throttle as the steering wheel comes back for the recovery. Often this will do the trick and you don't have to use throttle to save the oversteer.

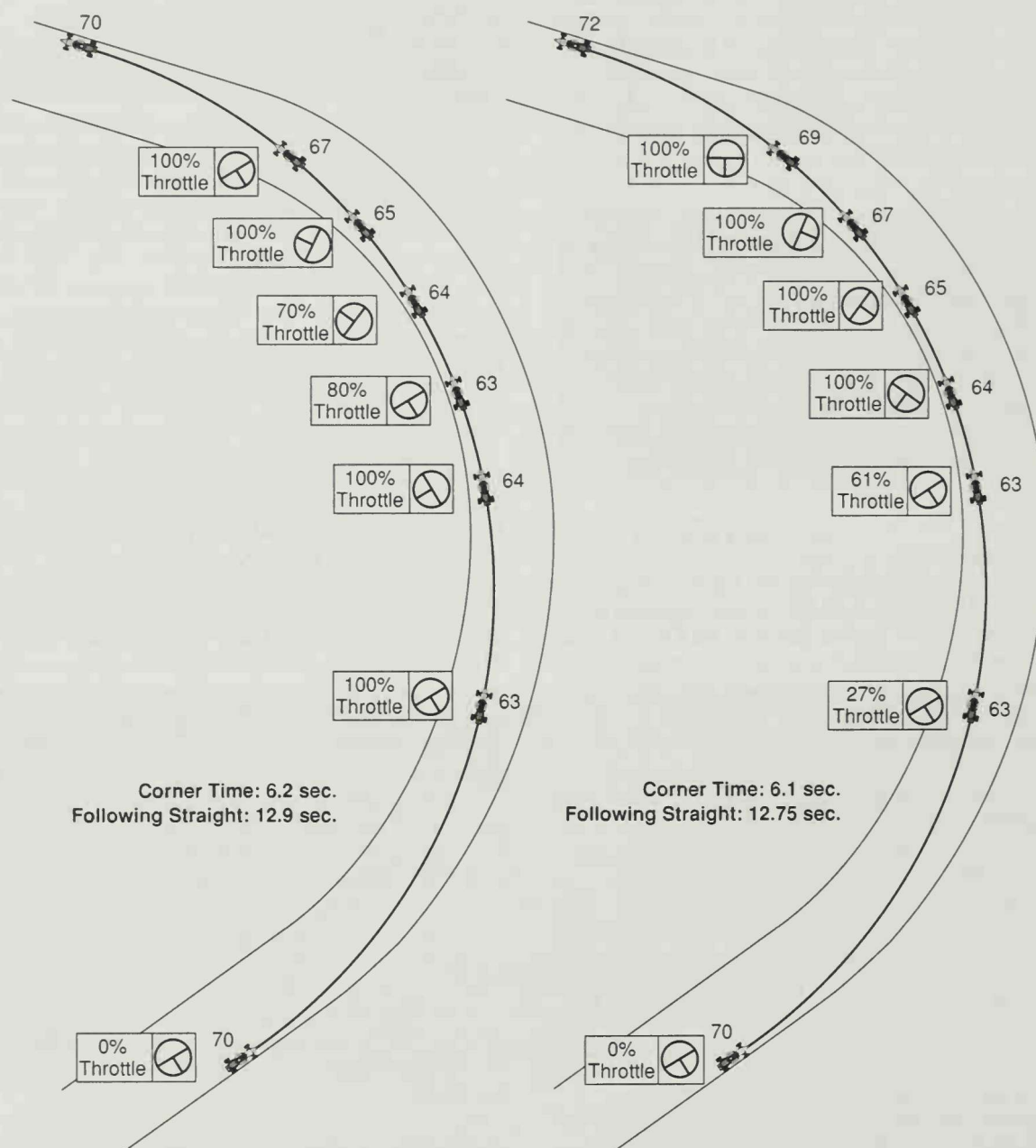


Fig. 4-9. Two approaches to Sebring's Carousel, the one on the left looking spectacular with big opposite lock corrections of power oversteer. The run on the right is more subtle and faster, both in the corner and down the following straight.

Add a Throttle Adjustment

In this particular case, though, the rear end pops out again. This time the countersteer goes past 90 degrees to the right. With a long straightaway to follow, you don't want to surrender any throttle, but at some point it becomes obvious that it's the only recourse.

You've got to be careful here. If you snap off the throttle, instead of restoring grip to the rear tires you would use even more of the rear tires' traction by engine braking. Combine this with unloading the rear tires by lifting, and it's a guaranteed spin. Happens all the time—a big rotation started by too much power, then taken beyond controllability by overreacting and snapping off the throttle.

With the correct amount of breathe, say, to about 70%, the rear tires get better bite, the rotation counterclockwise stops, the pause occurs, and you can start the recovery, getting the steering wheel back to straight, then to the left as the oversteer settles down.

This oversteer run through Turn 9 looked spectacular, but the elapsed time through the turn was a tenth of a second slower than a subsequent run where the throttle was fed in gradually but, once at 100%, stayed there.

More significantly, the interruption of throttle application reduced corner exit speed by 2 m.p.h., which hurt the elapsed time down the following straight and up to the entry to Turn 10 by another .15 seconds.

Comparing these two passes also shows you that what looks fast and spectacular is not necessarily the fastest way around. Especially coming onto the longest straight, drivers tend to get greedy about starting the application of power. You can see here that getting the power on early and aggressively, especially in a long corner, has its limits if it forces you to reduce power later in the corner to keep from spinning.

UNDER-ROTATION

As you learned earlier, understeer is an easier car control problem to solve than oversteer. Typically, understeer is caused by the front tires either being overused or underloaded.

The Overused Fronts

"Overused" means that the front pair of tires are often asked to do more of their share of the cornering of the car than the rears.

The majority of cars intended for street use are intentionally designed to understeer at their cornering limit. The thinking is that if an indifferent driver stumbles upon the car's cornering limits they are likely to lift off the throttle, which is exactly the right thing to do to restore grip to the front tires.

This setup is less than ideal for the racetrack, where you'd like the front and rear ends of the car to do their appropriate share of the direction change, allowing your car to get up into its ideal slip angle range.

Even with true racecars with their adjustable suspension systems, the cornering bias frequently ends up toward the front. Oversteer can be intimidating for racers as well as for street drivers, so the car is often adjusted to dial out the sensation of the tail of the car being further out than the nose.

What some drivers don't realize is that a neutral car (one where the front and rear tires are operating in the same range of slip angle) corners with the rear slightly further away from the inside of the corner than the front (refer back to Fig. 4-3). You shouldn't mistake this for oversteer. Many people do and, consequently, there are a lot of racecars out there that suffer from overused front tires and inherent understeer.

Underloaded Fronts

Even if the car does not have built-in understeer, you can provoke it by unloading the front tires: that is, using the controls in a way that reduces the download on the front tires.

The classic way to do this is to turn into the corner and simultaneously apply lots of throttle. If you do it right you can get the steering wheel turned to full lock and *still* not make the apex. Of course, doing this sort of thing routinely will generate tire temperatures off the gauge; the hotter they get, the worse they grip, the more they understeer, the hotter they get—on and on.

The trick is to turn into the corner with some load still on the front of the car, preferably by using the brake pedal past the turn-in point, trying to get the needed rotation up to the car's best slip angle.

In relatively fast corners where there isn't much speed loss required on the entry, the most common problem is getting understeer early in the turn. Two things seem to cause this problem most often. The first is that in a long fast corner, the throttle application point is typically closer to the turn-in than the apex. Drivers, in their never-ending attempt at maximizing exit speed, get greedy about putting the throttle down, unload the fronts and generate understeer.

The second cause is that aggressive throttle application, especially in a low powered car, doesn't have the same traction-robbing effect in higher gears as it does in lower gears. Experience has shown that in the higher gears, this early, abrupt throttle application more often results in unloading the front tires.

Sebring Chicane

A good place to look at this type of problem is in the second turn on the Sebring Test Circuit, the left hand swerve of the chicane. The chicane is made up of a third gear right, Turn 1, immediately followed by a left, Turn 2. You need to compromise the right—that is, take a later apex—so that at the exit of Turn 1 you're on the right side of the road, allowing a big radius through the left.

In our school race cars, you approach Turn 1 at the very top end of third gear, around 98 m.p.h. You brake

"After all these years, I was having trouble at Turn 6 in Miami and Roberto Lorrenzutti, who was driving with us, said, 'What a hard corner! you have to be so patient with the power—feeding, feeding, feeding, feeding.' I was trying to turn in and go immediately to lots of power, like a jerk. I teach this stuff every day of my life and still my right foot got the best of me. Next time out I tried just feeding it in, just short of wheelspin, and it was perfect, no correction, no late lifts, no nothing. So now, every corner I'm saying to myself 'feeding, feeding, feeding, feeding.'"

—Terry Earwood

down to 80 m.p.h. at the corner entry and get right back to full throttle through the apex, up to the turn-in point to the left, Turn 2. You swear you could turn in to Turn 2 without giving up full throttle, but if you do, the front end washes out into understeer and the apex becomes just a fantasy.

The technique that results in the quickest run through Turn 2 is to breathe the throttle just an instant before the turn to the left toward the apex, just enough to settle a little load onto the front tires. If you breathe the throttle back from 100% to 60% or 70% the car actually turns when you commit to the apex. Once it commits, you feed the power back on to full as you judge how much it will take without tripping the car back into understeer.

What's So Bad About Understeer?

A lot of drivers will say, "Wait a minute, I *like* understeer. The car's nice and stable. You're not constantly worried that the rear is going to take that big step out to uncontrollability. It gives the rear tires an additional measure of acceleration traction because you're using less of their cornering grip."

All true, but like everything else there's a compromise, depending on the degree of understeer. Subtle understeer may be overusing the fronts one percent more than what would be their fair share of the front/rear cornering balance. Overusing the fronts by 70% creates understeer of another magnitude, one that is so far off the optimum balance of the car that it will suffer from both a lack of total cornering grip and a resistance to acceleration due to tire scrub.

Each driver who fools with the cornering balance of the car can decide what balance is most comfortable. Some drivers will get very close to neutral, then bias things just slightly, that one percent, toward the front. Others will get very close to neutral and end up just a shade to the rear. Others, who are not so good at either setup or technique (or both) tend to end up *way* off of neutral, normally in the direction of understeer.

"Never Lift"

In some fast corners where the approach speed is almost the turn-in speed, you can get caught up in the pursuit of turning into the corner without giving up any throttle, trying to take the corner "flat out." The problem comes if turning in "flat" creates enough understeer that the car never makes the apex, or if the plowing of the front tires creates extra resistance to acceleration away from the corner. In these cases "flat out" might be slower than doing a slight lift to help the car rotate into the corner.

The "never lift" philosophy is frequently seen in drivers who generate understeer in a corner: everything seems okay, but then they spin at the corner exit. They "never lift" early in the corner where a breathe off of the throttle could get the car out of its understeer and help it turn more. Instead, they put in more and more steering lock, hoping that increasing the angle of the front wheels will get the nose of the car pointed down the following straight. Still, the right foot stays planted as the car understeers its way off the road at the track-out of the corner. *Then*, with dirt and rocks flying, you hear a big throttle lift and, as often as not, the car spins off the road to the inside.

"I've tried to get through Turn 1 at Mid Ohio flat out in a lot of cars and you can't quite do it, not faster than a lift anyway. You can turn in flat but then you have to give it up. If anything could do it would be a Super Vee and I tried it in one of those—couldn't do it."

—Robbie Buhl

Corner Exit Understeer

The "corner exit" is where you're trying to deliver as much power to the ground as you can, keeping the cornering speed up to maximize exit speed. Power oversteer can be a problem in this phase of the corner, and so can power understeer. It depends upon the car and situation. The same throttle application that produced oversteer in a high-powered, low-grip car will produce understeer in a high-grip, low-powered car.

It's impossible to make one statement that covers all cars. What you do know is how to solve the handling problem once you're in it: Oversteer is handled with the steering wheel by using the correction, pause and recovery technique; and by using throttle control to improve the bite of the rear tires.

Corner-exit understeer is corrected with throttle, but *not* steering. Adding more steering lock will most likely not deliver any more cornering ability to the front pair of tires. In fact, it will most often *reduce* the cornering force at the front by increasing the slip angle of the front tires into an area where they lose grip.

The answer is to lift off the throttle to keep from unloading the fronts as badly. Sometimes just a slight breathe is enough, sometimes you have to really come out of it, but beware of the snap off the throttle. With a violent lift you can turn a subtle understeer into a monster oversteer in a snap.

BIG ROTATION—PITCHING

Especially with experienced racers, it's a common notion that "pitching" the car early in a corner—that is, over-rotating it at the corner entry—allows you to do less cornering later in the turn. This is false. Doing more *cornering* early—steering the car on a tighter radius—is the only thing that allows you to do less cornering late in the turn where you would like to do more accelerating. Pitching, running the car at a higher yaw angle, only moves the tires away from their optimum slip angle.

Let's use a 90-degree corner as an example. The fundamental goal of driving through this corner is to change the direction that the car is travelling. In this case you have to change the car's direction 90 degrees.

Some would argue that if you rotate the car from zero degrees of yaw at the turn-in point to 8 degrees of yaw approaching the apex, you've reduced the amount of direction change needed by 8 degrees. But yaw and direction change are different.

Consider an extreme example (Fig. 4-10) where the driver has exercised brilliant car control and pitched the car sideways 90 degrees at the turn-in point. It's now pointed down the following straight-away, but this does not mean that all of the cornering is done. The driver still has to change the direction the car is *traveling* 90 degrees. The direction of travel (arrow A) is tangent to the arc the car is traveling on (arrow B) *regardless of the direction the nose is pointing*. Remember, it's the radius of the arc on which the car is traveling that determines how fast the car changes direction.

There are valid uses for big rotation. In corners that are very slow and of long duration, it doesn't hurt to rotate the car more than you might in a fast sweeper—especially if they have following straightaways. This rotation gives you a cushion against the car falling back into understeer, which will force you to get

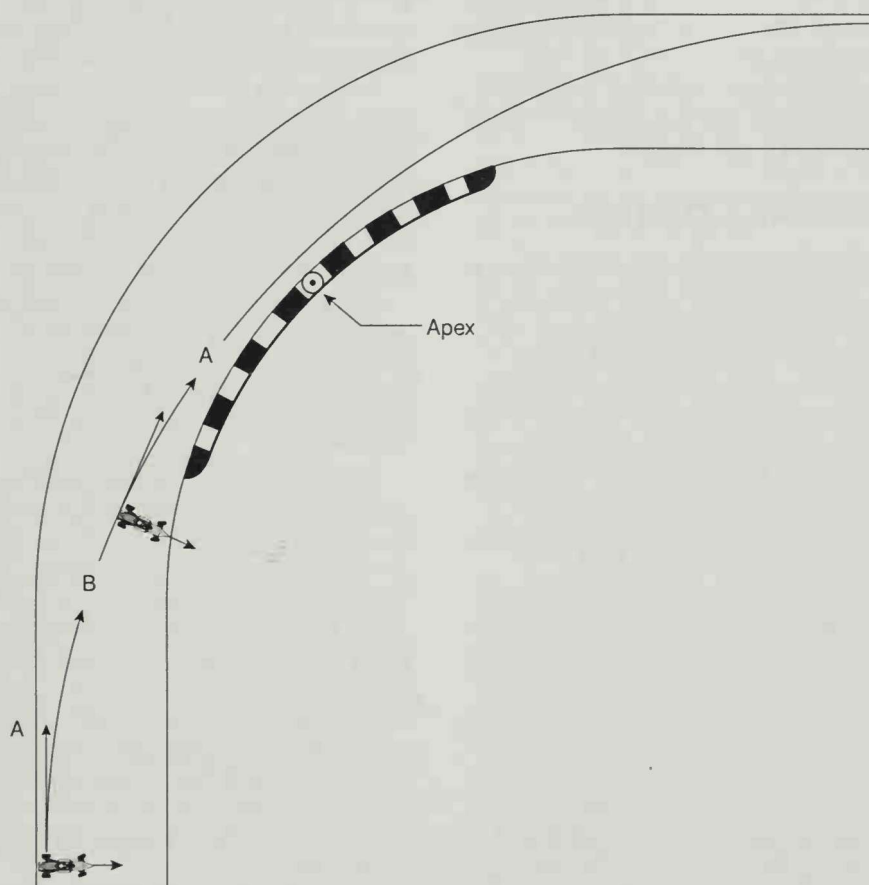


Fig. 4-10. "Pitching" the car seems like a valid way to change the car's direction rapidly, but on analysis the radius of the car's path relates to changes of direction, where yaw angle only represents the direction the nose is pointing.

out of the throttle to get the yaw angle back.

As cornering speeds get under 40 m.p.h. the rate of direction change per foot traveled increases rapidly. Tightening the radius of the first part of the corner by turning slightly later increases the rate of change of direction at double the rate that speed is sacrificed.

In looking at the data, it seems that pitching a car is less the reason that the approach is faster than an effect of the late turn-in—which really is the reason that acceleration can begin sooner, harder.

Generally, the cornering job is spread more uniformly across the corner as the corner gets bigger and faster. In slower corners, proportionately more cornering is done sooner in the turn than later, at least with the fastest drivers. As the turn gets tighter and slower, the spread between the amount of first-half cornering and second-half cornering gets bigger. In a fast sweeper, especially in an entry level car with limited horsepower, the balance of the change in direction between the first half and second half of the corner gets closer to 50/50.

Hand Brake Turns

Truly big rotation, often set off by using the handbrake to lock the rear wheels, can be useful in autocross, often to reverse the car's direction at the end of a slalom.

In this case the speeds are very low and a big, quick, rotation of 180 degrees, followed by acceleration essentially from a stop, can be faster than waiting for the car to turn through 180 degrees.

MULTIPLE-EVENT CORNERS

You will often have to deal with multiple problems in the same corner. Let's use Turn 7 at Sebring again for an example. On this particular lap you brake early, and at the turn-in point, the car is going 2 m.p.h. slower than normal. You turn in with very little rotation since the tires aren't at their limit. You get on the throttle at the same spot and with the same gusto as normal, and the car goes into understeer. You realize it and lift off the throttle. Weight transfers onto the front and off the rear, the car goes into trailing throttle oversteer and rotates up to an angle you like.

You stop the rotation with a touch of throttle and opposite lock. As the angle looks good, you continue the throttle's progress on toward the floor, keeping the slip angle within bounds. Just past the apex the road goes to reverse camber and the loss in traction causes the rear to slide wider. You react with opposite lock and a slight breathe off of full throttle to around 80%. The rotation stops, you recover the car and squeeze back down toward wide open. As the car moves toward the corner exit you gently unwind the steering wheel and accelerate at full throttle to the edge of the road at the track-out. You check the tach.

All of these moves on the steering wheel, throttle and brake are very typical of the way a car is driven at

its limit through a corner. The essence of car control is taking the car's cornering state at a particular instant and changing it to accomplish what you want—keeping the car in a range of yaw angle that lets the tires deliver the best grip on the road surface.

Despite the myths, car control is not a skill you're born with. The ability to keep a car accurately on the line as it's sliding through a corner requires that you know what to do and that you *practice* enough to develop the physical skills to do it.

PRACTICE

Professional schools offer programs that introduce you to the handling limits of street sedans. The lectures and exercises are invariably interesting and useful, but the skid pad is where the fun is. Most often, a car is intentionally driven in a circle at obscene yaw angles and the student has to practice the right moves to keep the car from spinning.

If you're considering a school of this type, ask about the nature of the skidpad or the skid car. If the pad is greased or the car you'll use is a specially built "skid car," the traction is so low that you won't experience the kind of loading sensations that you'll encounter in real life and, while the practice is useful, everything seems to happen in slow motion.

In our Skip Barber Dodge Driving Schools, which use a skidpad for part of the training, we simply wet down the skidpad to help the cars slide and to make the tires live for up to a day. The cornering forces are more realistic, and the sliding and recoveries occur over a time period which is more like real life. You're coached and encouraged to practice for hours.

The problem is that you can't afford a school every week of the year, and a couple days practice a year isn't enough to get good at any sport. Imagine practicing baseball two days a year and then expecting to be able to compete on a par with a Minor League player. Won't happen.

For practice, your racecar doesn't have to be a fancy, expensive one. Many road courses have open practice days, time trial events, or club races where the entry fees are low and the track time is abundant. Pick out the corners with the most runoff room so that if you make a mistake it would be hard to hit anything, then experiment with your limits. Slide the car more than you normally would in these corners and work on your car control. Don't just let things happen. Review the techniques you want to perfect and the car control problems you want to solve and work on them. Forget lap time and use each session as your own private clinic. It will pay off when you go racing in earnest.

Racing Karts is good practice, but just because they're small doesn't mean that they're cheap. Also realize that their response time and typical time spent in corners is significantly quicker than a full-size race-

"As far as spinning the car to find the limit, there are really a few ways to look at it. A driver has to learn just how far to go in cornering—you've got to know what the car can do. Although you don't want to lose control of the car, on the practical side, a driver without a lot of experience is going to overstep the bounds a few times. The thing is you have to pick your spots. There are places on each racetrack where there's a lot less risk if the car goes off and these are the places you'd play around with the limits with less risk of hurting the car or yourself.

"But once you're past the beginner stage you can't think that losing the car is OK. Two or three spins a weekend is way out of line. One every thirty days might be barely acceptable. The only use for the early spins is to learn from them—learn how not to do it again."

—David Loring

car. A lot has been learned about car control, inexpensively, in shopping mall parking lots after a snowfall. Don't tell the officer we sent you.

Using Your Eyes

You've heard the advice from your parents and maybe you even pass it on to the next generation: "Will you please watch where you're going!" It's good advice for a ten-year-old sprinting through the den, bouncing off end tables. It's even better advice for a racer.

When you're in a slide, it's very tempting to focus your eyes on the thing you're concerned about hitting—the oak tree, the lamp post, the AArmco barrier next to the racetrack. We've found that concentrating on looking where you want the car to go increases the probability of getting there. It's a matter of keeping the connection between the eyes and the hands.

From day one you develop the coordination to turn the steering wheel or handlebars just enough to get you where you're headed (where you're looking). When things get dicey and you're likely to act reflexively, you want your eyes focused on where you want to be, not on what you want to avoid. In racing you want your eyes on the apex or the track-out and your hands will tend to move the steering wheel to get you there. It sounds like a minor hint but looking where you want to go is a significant aid for finding the racing line and staying on it, even when the car is sliding wildly.

"The eyes are the single most important partner in driving the race car. If we were writing a book on baseball, the first sentence would be, 'The most critical thing you can do is to keep your eye on the ball.' In the sport of driving, we're sitting in the ball. We should be looking where we want to go....

"The first time you read that sentence, you think, 'Of course I'm looking where I want to go,' but it's not guaranteed—you slide the car and you tend to look where the car is sliding, not where you want it to end up. When you come up to a corner, it's easy to stare at the turn-in point and not look into the corner soon enough. We think we look where we want to go but often we don't unless we concentrate on it."

—Terry Earwood

FRONT/ALL WHEEL DRIVE

With the advent of showroom stock racing in the 1970s, more and more front wheel drive cars were showing up at racetracks. With more than a decade of road racing experience you still don't see front wheel drive being employed by many racecars that are built specifically for racing. Teams race them if they must, but there is no competitive advantage in the front wheel drive configuration.

The problem with front wheel drive is that the driven wheels become unloaded under acceleration, reducing the tires' traction just at the time you need it. As a general rule, front wheel drive cars create power understeer in circumstances where a conventional drive car creates power oversteer. Where an aggressive burst of power robs traction from the rear of the car in a conventional setup, the same burst robs cornering grip from the front of a FWD car. This traction loss is worse than it is in a conventional layout.

If you were starting from scratch designing a racecar, you would much rather have the driving wheels on the end of the car that will gain tire loading under acceleration: the rear pair of tires. Nonetheless, if you're racing showroom stock and your car of choice happens to have front wheel drive, you have to learn to deal with it.

Braking Control

A properly biased braking system is even more important on a FWD car. Many FWD cars which start out with a small percentage of their weight at the rear of the car will transfer almost all of it forward under braking. Once the car is turned into the corner and

load transfers off the inside tires to the outside tires, you may end up with the inside rear tire in the air.

You're now cornering with the two front tires and the outside rear. If there's a little too much rear brake bias, the rear end will swing wide, creating substantial oversteer on corner entry.

Just like RWD, the front drive car will go toward oversteer if you snap off the brake pedal while cornering. The fronts, relieved of their braking requirements and loaded heavily from weight transfer, will dart toward the inside; the rears, high in the air and unloaded, will slide toward the outside of the turn.

The pause phase of trail braking that can help rotate a RWD car by allowing engine braking to help the rear end slide doesn't affect the FWD driver. You can still use the pause as a time to wait for the rear to reach the yaw angle that you like before stabilizing it with throttle. FWD doesn't have the tendency toward oversteer that engine drag on the rear tires gives to the RWD car.

Throttle Control

Brake bias-induced oversteer is not altogether a bad thing. If the rear comes out to even a quite lurid angle, the built-in understeer that will take over once you apply the throttle will bring the car back toward neutral.

In a FWD car, oversteer is solved with the same correction, pause, recovery routine used in a RWD car, but it also responds eagerly to throttle application. In the RWD car, you had to be careful about not overdoing the throttle application, losing rear tire traction and making the oversteer worse. In FWD, adding power and opposite lock drives the front end of the car over in front of the rear. If you overdo it and add too much power, the front loses traction and slides out onto a bigger arc, reducing the angle of the oversteer. Either way you win by adding power and opposite lock in a FWD oversteer slide.

Like the rear wheel drive car the FWD tends to create understeer with gradual throttle application. Again, just like the RWD, lifts off the throttle will change the handling of the car away from understeer and toward oversteer.

We haven't experimented with all-wheel drive on the racetrack. Generally, they share a greater similari-

ty with FWD cars than rear drivers. They tend to create more power understeer than conventional racecars. One common comment by racers who frequently drive all wheel drive cars is that, in rain races, the additional traction from two more driving wheels is a huge advantage.

SUMMING UP

The role of car control is to keep the car on your intended path at a yaw angle in the range that creates the best grip from the tires, increasing the speed with which you can negotiate the cornering arc.

Controlling the car from the turn-in point of the corner to the throttle application point involves mixing braking loads with cornering loads. Gradually reducing braking loads will tend to maintain the car's steady state cornering balance: oversteer will stay oversteer, understeer will stay understeer and neutral will tend to stay neutral.

Suddenly popping off the brake pedal while cornering will move the car's handling toward oversteer. A pause between brake release and throttle application will tend to move the car toward oversteer.

Gradual increases in throttle moves the car in the direction of understeer. Abrupt throttle applications move the car in the direction of oversteer. Lifts off the throttle change the car's balance toward oversteer.

Correction, pause and recovery are the three phases of the steering solution to oversteer skids and slides.

Rotation is the process of the car going from zero yaw to the cornering yaw angle necessary to develop the optimum slip angles and thereby the ultimate grip of the tires. Even "neutral" handling cars develop yaw angles.

Looking in the direction you would like the car to go can help by using your developing coordination between your eyes and extremities.

Car control is not an inborn skill any more than hitting a golf ball is inborn. Understanding what you want to do, and practicing the physical moves required, develops car control.